

FINAL Report

Digital Theremin

a Full Implementation with Renesas MCU AIK-RA8D1

Team Lemon Tea

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Abstract

Theremin is an analog instrument that is notorious for its expensive price and being hard to perform. Furthermore, the analog structure makes it harder to incorporate with the nowadays Digital Audio Workstation.

I. INTRODUCTION

I believe that our group is one of the only few groups giving a hoot about the assigned Renesas MCU as the main research focus.

II. CIRCUIT DEPLOYMENT

Our implementation includes three main parts.

- 1 AIK-RA8D1 Development Board The core MCU of this experiment, responsible for control, signal acquisition, and data processing. Provides ADC pins (e.g., AN121, AN000, AN001) for analog signal sampling and transfers data to PC via USB CDC. Supports interfaces such as I²C and UART for peripheral communication.
- 2 HC-SR04 Ultrasonic Module Transmitter (Tx): Triggered by a high-level signal ($<10\ \mu\text{s}$), it emits 8 pulses at 40 kHz. Receiver (Rx): Captures the reflected signal, which is sampled by the MCU's ADC.
- 3 HP-122 Passive Buzzer
Distance is calculated through time-of-flight (ToF) measurement, considering the speed of sound in air $V=331+0.6t\ \text{m/s}$.

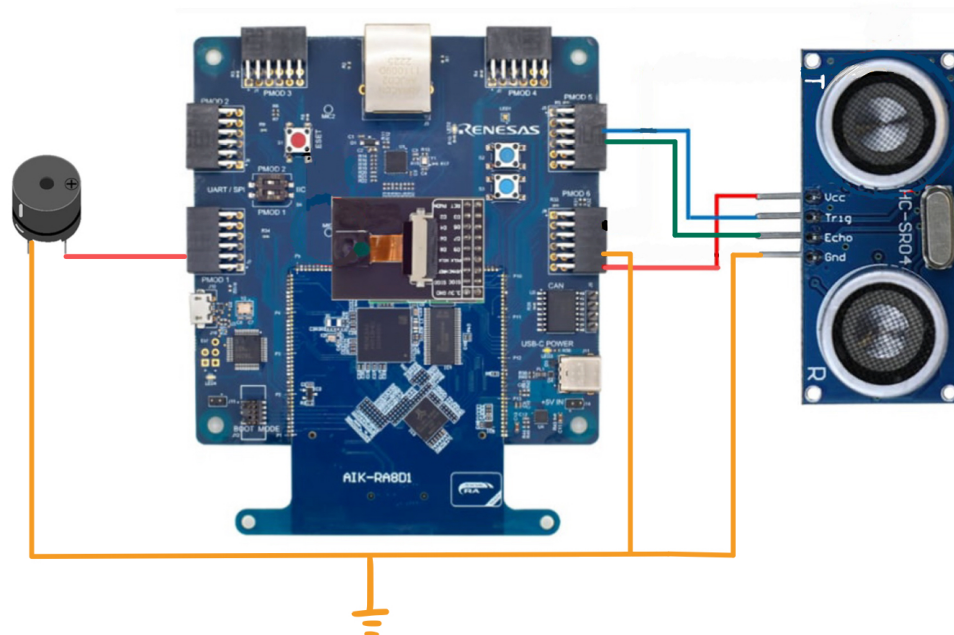
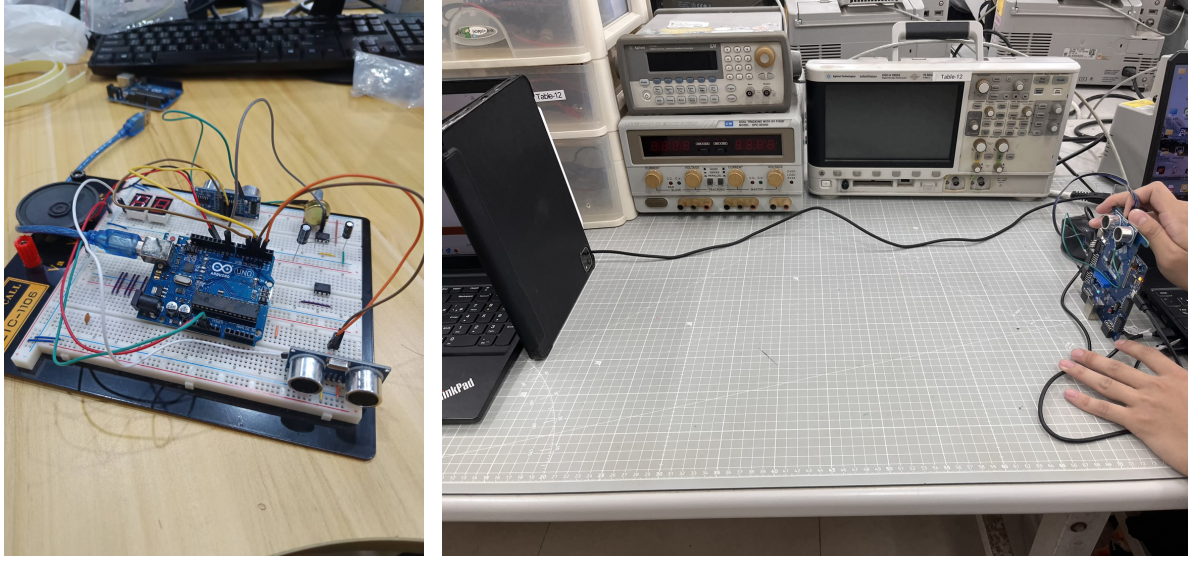


Fig. 1: hardware configurations



(a) Circuit deployment on our beloved AIK-RA8D1. (b) measurement experiments setups. We use our tablet as the reflection board, and the distance is referenced with the table pad.

Fig. 2: hardware configurations

III. FPS CONFIGURATIONS

IV. CALLBACK SUBMODULES IMPLEMENTATIONS

V. MAIN CONTROL FLOW IMPLEMENTATION

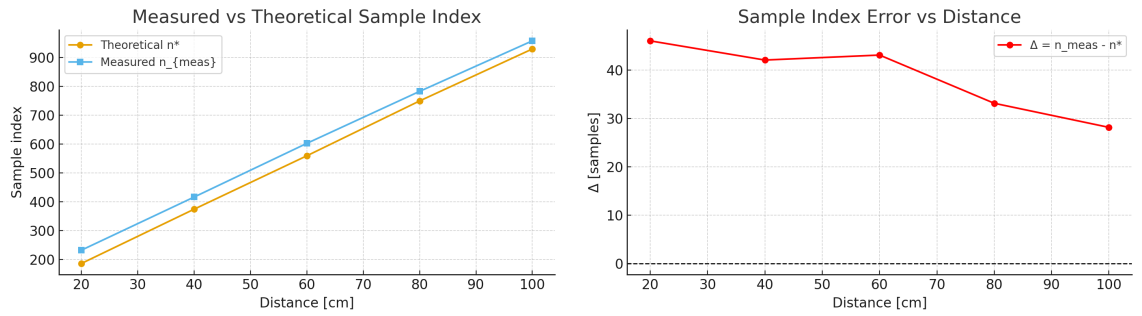
VI. ERROR SOURCE ANALYSIS

We analyzed the measured peak indices across multiple distances and compared them with the theoretical sample indices n^* computed from the speed of sound at $T = 25^\circ\text{C}$. This allowed us to examine the trend of the measurement error and construct a model for calibration.

Pair (cm)	n_{theory}^*	n_{meas}	$\Delta n_{\text{theory}}^*$	Δn_{meas}	$\frac{\Delta n_{\text{meas}}}{\Delta n_{\text{theory}}^*}$	Relative Error (%)
20→40	184.97→369.94	231→412	184.97	181	0.979	-2.1
40→60	369.94→554.91	412→598	184.97	186	1.006	+0.6
60→80	554.91→739.88	598→773	184.97	175	0.946	-5.4
80→100	739.88→924.86	773→953	184.97	180	0.973	-2.7

TABLE I: Relative Distance Comparison (Measured vs. Theoretical)

The measured peak indices n_{meas} were aligned relative to the start of transmission, determined by the maximum slope of the Tx waveform. We then plotted n_{meas} against n^* , as well as the corresponding sample index error $\Delta n = n_{\text{meas}} - n^*$, as shown below.



(a) Measured peak index and theoretical n^* at $T = 25^\circ\text{C}$

(b) Sample index error vs. distance

Fig. 3: Comparison between measured peak indices and theoretical sample indices.

From Fig. 3b, we observe that the error Δn decreases approximately linearly as distance increases, consistent with a fixed system delay plus a small distance-dependent effect. Instead of the usual choice of a constant model, we found out that the error is related to the distance. We therefore model the error as

$$\Delta n = a + b n_{\text{meas}},$$

and use linear regression to estimate $a \simeq 53.14$ and $b \simeq -0.02469$. Applying this model, we compute corrected sample indices $\hat{n} = n_{\text{meas}} - \Delta n$, yielding the results summarized below.

True Distance (cm)	n_{meas}	$\hat{n}$ (Corrected)	n^* (Theory)	Error After Correction (cm)
20	231	183.56	184.97	-0.15
40	412	369.04	369.94	-0.10
60	598	559.39	554.91	+0.48
80	773	738.65	739.88	-0.13
100	953	922.23	924.86	-0.28

TABLE II: Corrected Sample Index and Residual Distance Error (Linear Model)

The corrected indices significantly reduce the bias, with all residual distance errors falling within ± 0.5 cm, demonstrating that the linear error model is effective.

To translate the measured sample index back into a physical distance, we use the relation between the sample index n , the sampling frequency F_s , and the speed of sound v . Each sample corresponds to a time increment of

$$T_s = \frac{1}{F_s},$$

so the round-trip time corresponding to n samples is

$$t = n T_s = \frac{n}{F_s}.$$

Assuming a two-way (Tx-Rx) propagation path, the one-way distance is then given by

$$\hat{d} = \frac{v t}{2} = \frac{v}{2 F_s} n.$$

For our experiment at $T = 25^\circ\text{C}$, we use $v = 331 + 0.6T \approx 346$ m/s = 34,600 cm/s and $F_s = 160$ kHz, which yields the conversion factor

$$k = \frac{v}{2 F_s} = \frac{34,600}{2 \times 160,000} \simeq 0.1081 \frac{\text{cm}}{\text{sample}}.$$

Thus, the estimated distance from a measured sample index n_{meas} (or its corrected version $\hat{n}$) is

$$\hat{d} = k n.$$

This linear relation allows us to directly convert the corrected sample indices from Table II into absolute distance estimates and to compute residual errors in centimeters for performance evaluation.

To fully exploit our calibration, we convert our measured width to corresponding number of samples using $F_s = 160\text{kHz}$

distance(cm)	width(μs)	sample(n)	$\hat{d}(\text{cm})$
5	327.5	52	5.66
10	620.8	99	10.74
15	941.4	151	16.28
20	1205.5	193	20.85
23	1240.2	198	21.45
30	1848.9	296	31.97

TABLE III: Relative Distance Comparison (Measured vs. Theoretical)

VII. ACKNOWLEDGEMENT

We're very appreciated all the support with the Renesas Team, especially the engineers Steven and Henry, and thanks to Hedy for the effective and responsive communication and schedule coordination. Also, thanks for all the TAs for your

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